and also "$ac = 54$".

$\therefore$ we split the middle term as $6x$ and $9x$

$$\begin{aligned} 2x^2 + 15x + 27 &= 2x^2 + 6x + 9x + 27 \\ &= 2x(x+3) + 9(x+3) \\ &= (x+3)(2x+9) \end{aligned}$$

Therefore, $(x+3)$ and $(2x+9)$ are the factors of $2x^2 + 15x + 27$.

Example 3.28

Factorise $2x^2 - 15x + 27$

Solution

Compare with $ax^2 + bx + c$

$a = 2,\ b = -15,\ c = 27$

product $ac = 2 \times 27 = 54$, sum $b = -15$

$\therefore$ we split the middle term as $-6x$ and $-9x$

$$\begin{aligned} 2x^2 - 15x + 27 &= 2x^2 - 6x - 9x + 27 \\ &= 2x(x-3) - 9(x-3) \\ &= (x-3)(2x-9) \end{aligned}$$

Therefore, $(x-3)$ and $(2x-9)$ are the factors of $2x^2 - 15x + 27$.

Product of numbers $ac = 54$	Sum of numbers $b = -15$	Product of numbers $ac = 54$	Sum of numbers $b = -15$
1×54	55	-1×-54	-55
2×27	29	-2×-27	-29
3×18	21	-3×-18	-21
6×9	15	**-6×-9**	**-15**
The required factors are **-6** and **-9**			

Example 3.29

Factorise $2x^2 + 15x - 27$

Solution

Compare with $ax^2 + bx + c$

Here, $a = 2,\ b = 15,\ c = -27$

product $ac = 2 \times -27 = -54$, sum $b = 15$

$\therefore$ we split the middle term as $18x$ and $-3x$

$$\begin{aligned} 2x^2 + 15x - 27 &= 2x^2 + 18x - 3x - 27 \\ &= 2x(x+9) - 3(x+9) \\ &= (x+9)(2x-3) \end{aligned}$$

Therefore, $(x+9)$ and $(2x-3)$ are the factors of $2x^2 + 15x - 27$.

Product of numbers $ac = -54$	Sum of numbers $b = 15$	Product of numbers $ac = -54$	Sum of numbers $b = 15$
-1×54	53	1×-54	-53
-2×27	25	2×-27	-25
-3×18	**15**	3×-18	-15
-6×9	3	6×-9	-3
The required factors are **-3** and **18**			

Example 3.30

Factorise $2x^2 - 15x - 27$

Solution

Compare with $ax^2 + bx + c$

Here, $a = 2,\ b = -15,\ c = -27$

product $ac = 2 \times -27 = -54$, sum $b = -15$

$\therefore$ we split the middle term as $-18x$ and $3x$

Product of numbers $ac = -54$	Sum of numbers $b = -15$	Product of numbers $ac = -54$	Sum of numbers $b = -15$
-1×54	53	1×-54	-53
-2×27	25	2×-27	-25
-3×18	15	**3×-18**	**-15**
-6×9	3	6×-9	-3
The required factors are **3** and **-18**			

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